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Market Research SMR (Small Micro Reactor) RFI

THIS IS NOT A SOLICITATION FOR PROPOSALS NOR DOES IT COMMIT THE GOVERNMENT TO CONTRACT FOR SUPPLIES OR SERVICES.

Notice: In accordance with FAR 52.215-3,

- (a) The Government does not intend to award a contract on the basis of this solicitation or to otherwise pay for the information solicited.
- (b) Although "proposal" and "Offeror" are used in this Request for Information, your response will be treated as information only. It shall not be used as a proposal.
- (c) This solicitation is a "Sources Sought" Synopsis to determine potential sources for information and planning purposes only. The purpose of this Sources Sought is to conduct market research to determine if responsible sources exist, to assist in determining if this effort can be competitive and/or a total Small Business Set-Aside. The Government will use this information to determine the best acquisition strategy for this procurement. All information received in response to this RFI that is marked PROPRIETARY will be handled accordingly. The Government shall not be liable for or suffer any consequential damages for any proprietary information not properly identified. Proprietary information will be safeguarded in accordance with applicable Government regulations. Please be advised that all submissions become Government property and will not be returned.

Agency: Department of the Air Force

Contracting Office: AFTC/PZK

Pre-Solicitation Number: FA8240-26-R-3001

NAICS: 221113

Response Date: NLT 19 April 2026, 4:00 PM MDT

Objective: The purpose of this Request for Information (RFI) is to assist the Government in identifying potential offerors for understanding the company's capability to design, license, fuel, construct, and deploy Small, Micro, or Modular Reactor (SMR) technologies in compliance with applicable regulatory, safety, environmental, and security requirements. The information collected will help assess industry confidence in achieving and installing a commercially viable Nth-of-a-Kind (NOAK) capability.

Background: The Government is conducting market research to better understand the current state of Small, Micro, or Modular Reactor (SMR/MMR) technologies and the industrial base supporting their development and deployment. Advances in reactor design, fuel innovation, manufacturing, regulatory pathways, and construction methodologies have created potential opportunities for scalable, resilient, and reliable energy solutions. As part of this effort, the Government seeks information regarding industry capabilities to support compliant, fuel-secure, and commercially scalable reactor deployment strategies.

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If interested in this opportunity, please include the following information with your response to this RFI:

- Full Company Name:
- CAGE Code:
- Type of Business (small/large):
- POC for this effort:
- POC Email:
- POC Phone:
- Currently registered in System of Acquisition Management (SAM): Yes / No
- Does your company have an approved Facility Clearance Level (FCL) by the Defense Counterintelligence & Security Agency (DCSA)? Yes / No. If yes, what level?
- Does your company currently have, or has it ever been issued a DD Form 254, DoD Contract Security Classification Specification, allowing work using classified information? Yes / No. If yes, what level/s? Please provide narrative.
- Provide a description of similar services offered to the Government and/or to commercial customers during the past three years. Any responses involving teaming agreements should delineate between the work that will be accomplished by the prime and the work accomplished by the teaming partners.

1. Corporate Profile & Industrial Base

1. Describe your company's core SMR technology offering, including reactor type, output range, and deployment model.
2. Identify your position within the nuclear value chain (e.g., designer, integrator, constructor, operator).
3. Describe your domestic manufacturing footprint and supply chain structure.
4. Identify key critical suppliers (long-lead items, fuel, HALEU, specialized forgings, etc.) and note any foreign dependencies.
5. What are the leading indicators of failure?
6. What early warning signals should be monitored?

2. Technical Capability & Technology Maturity

7. State the current Technology Readiness Level (TRL) of your proposed reactor technology.
8. Describe any prototype, test reactor, or demonstration activities completed or underway.
9. Provide estimated timeline to first deployment (FOAK) and subsequent NOAK deployment.
10. Describe design features that enable modularity, transportability, or rapid installation.
11. Identify key technical risks that could delay deployment and your mitigation strategies.
12. What are the known failure modes and corresponding mitigation strategies?
13. How would reactor downtime impact mission operations, and what redundancy is designed?
14. Describe the load requirements for testing and/or operational use.

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15. How would the system status and operations be monitored?
16. Would a remote connection (off the serviced installation) be required for system monitoring or operations?

3. Regulatory & Licensing Readiness

17. Identify current licensing status (e.g., pre-application engagement, submitted application, approved design).
 18. Estimate regulatory timeline under realistic assumptions.
 19. Identify regulatory steps outside your direct control.
 20. Describe your approach to environmental compliance (NEPA, state permitting, etc.).
 21. What is the realistic NRC (or other regulatory) licensing timeline/pathway, including best-case, expected, and worst-case scenarios?
 22. What regulatory precedents exist for co-located classified infrastructure and nuclear facilities?
 23. What regulatory outcomes would result in project termination?
 24. What regulatory inspections and oversight are required or expected for ongoing operations?
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4. Fuel Supply & Fuel Cycle Strategy

25. Specify fuel type, enrichment level, and burn cycle.
 26. Identify fuel qualification status.
 27. Describe your fuel supply chain strategy, including domestic sourcing.
 28. Identify risks associated with HALEU or other enrichment supply constraints.
 29. Describe spent fuel handling, storage, and disposition approach.
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5. Deployment & Construction Strategy

30. Describe your construction methodology (stick-built, modular fabrication, factory-built).
 31. Provide estimated installation timeline from site readiness to initial criticality.
 32. Identify site infrastructure requirements (water, security perimeter, grid interconnection).
 33. Describe workforce requirements and availability.
 34. Explain how your design supports repeatable NOAK deployment at reduced cost and schedule.
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6. Safety, Security & Compliance

35. Describe inherent safety characteristics of the design.

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36. Outline physical security, cyber security, and insider threat mitigation approaches.
 37. Describe emergency preparedness framework and coordination requirements.
 38. Explain nuclear quality assurance (QA) and safety culture governance structure.
 39. Identify any safety or licensing issues that could materially impact viability.
 40. What is the emergency response concept of operations?
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7. Financial & Commercial Structure (High-Level)

41. Provide high-level capital cost range (FOAK vs. NOAK if available).
 42. Identify major cost drivers.
 43. Describe allocation of construction and performance risk.
 44. How is nuclear liability insured and structured?
 45. Who holds long-term decommissioning and waste disposition responsibility?
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8. Mission Integration & Scalability

46. Describe how your system supports mission resilience (islanding capability, black start, load following, redundancy).
47. Explain scalability across different installation sizes.
48. Identify conditions under which deployment would not be technically or economically viable.
49. Describe long-term operations and maintenance model.
50. Identify the top three risks to achieving a commercially viable NOAK installation and proposed mitigation strategies.

9. Execution & Organizational Capability

51. What, if any, nuclear projects of similar scope have the team executed?
52. Who serves as the prime nuclear integrator?
53. How will nuclear safety culture, QA, and compliance be governed?
54. What organizational controls ensure regulatory adherence?
55. Recommend contract structures appropriate for this type of effort.
56. Identify potential barriers to competition within the current industrial base.
57. Provide any additional information that would assist the Government in determining acquisition strategy.

Please provide any questions or recommendations you feel are applicable to assisting with the SMR Program market research.

An approved Facility Clearance Level (FCL) and employee clearances will be required to participate in the solicitation, when issued.

Please email responses to this RFI to the Government POCs listed below.

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Government Points of Contact:

- Samuel Christensen, Contracting Officer, samuel.christensen.1@us.af.mil
- Rochelle Reynolds, Contracting Specialist, rochelle.reynolds.1@us.af.mil
- David Dammeier, Col, USAF, Project Lead, NTTR/MS
- Jose “Javi” Mendez, Deputy Director, NTTR/MS